

SDM_FREEZE_FORENSIC_REPORT

Framework: Strategic Constitution Audit Framework (SCAF) **Version:** 1.0 **Target:** SDM_V2.1.md

Auditor: Constitutional Forensic Adjudicator

AUDIT TARGET 01 — Research Authority vs Trade Authority

Finding F-01A | Advisory Boundary Definition

Status: REJECTED **Severity:** N/A **Blocking:** NON-BLOCKING

The constitution defines the advisory boundary with clarity and redundancy. SDM-CONST-01, SDM-CONST-06, SDM-CONST-13, SDM-01, SDM-10, GOV-01, and GOV-02 collectively and consistently establish that all system outputs are advisory, that execution authority belongs exclusively to the human, and that the system governs recommendation authority only. SADR generation requires no assumption here.

Finding F-01B | System Activities Enumeration

Status: REJECTED **Severity:** N/A **Blocking:** NON-BLOCKING

SDM-CONST-01 explicitly enumerates the five activities the system is authorized to perform (identify, rank, recommend, provide evidence, suggest exits) and three it is not (autonomous trading, execution bot, HFT/day trading). The distinction is constitutionally complete for SADR purposes.

AUDIT TARGET 02 — HALT Semantics

Finding F-02A | Halt Term Proliferation

Status: CONFIRMED **Severity:** MEDIUM **Blocking:** BLOCKING

The constitution uses four distinct halt-class terms without establishing a unified semantic taxonomy:

1. **Hard Halt** — SDM-15 Rule 5: "Position limit breaches must trigger hard deterministic halts rather than passive alerts."
2. **Governance Halt** — GOV-01 Rule 3–5: Triggered by portfolio drawdown breach. Suspends recommendations. Requires human authorization to exit.

3. **Governance Lockout** — GOV-02 Rule 1-3: Triggered by human risk governance violation. Suspends recommendations. Exits on compliance restoration.
4. **Halt (generic)** — SDM-15 Rules 6, 7, 8, 11, 12: Circuit breaker halts on recommendations during specific market conditions.

SADR Impact: SADR generation must understand whether these are four distinct system states, whether they compose (can Governance Halt and Governance Lockout be active simultaneously), and what the exit condition is for each generic halt in SDM-15. Without a taxonomy, architecture teams will invent halt composition rules and exit semantics. This is a blocking ambiguity.

Layer 2: BLOCKING **Layer 3:** SAFE_TO_RESOLVE — no owner intent is modified by clarifying that halt terms are distinct non-overlapping states.

Finding F-02B | Governance Halt Exit Condition Specificity

Status: CONFIRMED **Severity:** LOW **Blocking:** NON-BLOCKING

GOV-01 Rule 5 states "Normal operation shall not resume until the human has reviewed and authorized resumption." The exit condition is human-authorized resumption. This is constitutionally sufficient. The precise mechanism of authorization is an implementation concern, not a constitutional gap. SADR generation can work from this definition without inventing behavior.

Finding F-02C | Governance Lockout Exit Condition — "Compliance Restored"

Status: CONFIRMED **Severity:** MEDIUM **Blocking:** BLOCKING

GOV-02 Rule 3 states "Governance Lockout remains active until governance compliance is restored." The term "governance compliance restored" is not defined. A human may restore compliance by: (a) re-instating the stop-loss, (b) bringing the position into approved risk bounds, (c) explicitly acknowledging the violation, or (d) some combination. SADR generation cannot determine which of these constitutes restoration without inventing a definition.

Layer 2: BLOCKING **Layer 3:** SAFE_TO_RESOLVE — the owner's intent (behavioral circuit breaker) is clear; clarifying the exit condition does not modify that intent.

AUDIT TARGET 03 — Report Generation Semantics

Finding F-03A | Report Cadence Undefined

Status: CONFIRMED **Severity:** MEDIUM **Blocking:** BLOCKING

The constitution mandates that reports must be produced (SDM-CONST-11, SDM-12, SDM-08, SDM-09, SDM-10) and specifies what content they must contain. However, the constitution does not define when reports are generated. Two fundamentally different models exist:

- **Model A — On-Demand:** The human requests analysis. The system responds.
- **Model B — Scheduled:** The system generates reports on a scheduled cadence (e.g., daily pre-market).

These models produce different architectural and data-dependency requirements. SADR generation cannot select between them without inventing a decision that belongs to the owner.

Layer 2: BLOCKING **Layer 3:** OWNER_DECISION_REQUIRED — report cadence directly affects how the system is invoked, which touches ODP-001 (system purpose) and ODP-014 (modular design). This cannot be resolved by constitutional inference.

Amendment: NOT CREATED. Owner decision required.

Finding F-03B | Report Production Responsibility

Status: REJECTED **Severity:** N/A **Blocking:** NON-BLOCKING

The constitution consistently assigns report production to the system (SDM-CONST-11, SDM-08 Human Visibility, SDM-10 Human Visibility). No ambiguity exists about which entity produces reports. SADR generation requires no assumption.

Finding F-03C | Governance Halt — Critical Risk Escalation Report

Status: CONFIRMED **Severity:** LOW **Blocking:** NON-BLOCKING

GOV-01 Rule 4 mandates "a critical risk escalation report shall be generated for human review" upon entering Governance Halt mode. The report is event-triggered (drawdown breach) and its purpose is human review before resumption. The content standard follows SDM-CONST-11 (risk summary) and SDM-15 (human visibility: drawdown versus limit, halted states). SADR generation can derive sufficient behavior from existing constitutional content. Non-blocking.

AUDIT TARGET 04 — Hidden Ambiguity Discovery

Finding F-04A | SDM-15 Generic Halt — No Exit Condition

Status: CONFIRMED **Severity:** MEDIUM **Blocking:** BLOCKING

SDM-15 Rules 6, 7, 8, 11, and 12 mandate that the system shall "halt recommendations" under specific conditions (hedging cycles, uncertainty band widening, volume spikes, macro shocks, non-ergodic breakdowns). None of these halt conditions define an exit state. A halt with no defined exit is an indefinite operational suspension. SADR generation will be forced to invent re-entry conditions for each.

Layer 2: BLOCKING **Layer 3:** SAFE_TO_RESOLVE — the owner's intent is recommendation suspension during adverse conditions, with resumption when conditions normalize. Clarifying the exit principle does not modify intent.

Finding F-04B | "Hard Deterministic Halt" in SDM-15 Rule 5 — Unresolved Distinction from Governance Halt

Status: CONFIRMED **Severity:** LOW **Blocking:** NON-BLOCKING (subsumed by F-02A)

This finding is absorbed into F-02A (halt taxonomy). The term "hard deterministic halt" in SDM-15 Rule 5 (position limit breaches) requires clarification as part of the halt taxonomy amendment. No separate amendment is needed.

Finding F-04C | Opportunity Ranking "Open Menu" — No Expiry Defined

Status: CONFIRMED **Severity:** LOW **Blocking:** NON-BLOCKING

SDM-08 Rule 8 establishes that the human may select any ranked opportunity from the presented menu in any order. The constitution does not define how long the menu remains valid (i.e., whether opportunities selected 24 hours after ranking are still constitutionally authorized). This is a data-freshness question, not a constitutional governance question. It is governed implicitly by the continuous evidence assessment requirement in SDM-12 and the validation requirement in SDM-05. Non-blocking for SADR.

Finding F-04D | Two Distinct Halt Exit Authorities

Status: CONFIRMED **Severity:** LOW **Blocking:** NON-BLOCKING

Governance Halt exits on explicit human authorization (GOV-01 Rule 5). Governance Lockout exits on "compliance restored" (GOV-02 Rule 3, per F-02C amendment). These are constitutionally appropriate asymmetric exits — one is human-authorized, one is condition-based — consistent with their distinct triggers. No contradiction exists. Finding absorbed into F-02A taxonomy amendment and F-02C amendment.

Finding F-04E | SDM-15 Rule 3 — "Execution Logic Must Remain Deterministic"

Status: REJECTED **Severity:** N/A **Blocking:** NON-BLOCKING

The phrase "execution logic" in SDM-15 Rule 3 ("Execution logic must remain strictly deterministic. The AI is advisory and must never dictate execution sizing or routing.") might appear to imply system execution authority. In constitutional context, this rule constrains the AI's advisory output to be deterministic rather than probabilistically variable. It does not grant execution authority. The sentence that follows ("The AI is advisory and must never dictate execution sizing or routing") confirms this reading. No ambiguity for SADR purposes.

Finding F-04F | "Case-by-Case" Evaluation Protocol — Undefined

Status: CONFIRMED **Severity:** LOW **Blocking:** NON-BLOCKING

SDM-10 Rule 4 states that when disagreements occur between the system recommendation and the owner's decision, "a case-by-case evaluation protocol shall be triggered." The protocol itself is not defined. However, this is constitutionally appropriate: it is a discretionary human process, not a system process. The system's obligation is to trigger it (i.e., log the disagreement and present the conflict). Architecture does not need to define the human deliberation process. Non-blocking.

Finding F-04G | Conviction Weighting Upper Bound — No Maximum Concentration Value

Status: CONFIRMED **Severity:** LOW **Blocking:** NON-BLOCKING

SDM-07, SDM-08, and SDM-09 collectively establish conviction-weighted allocation with a maximum concentration constraint. The specific maximum concentration percentage is not

constitutionally defined and appears in the [VALIDATION_REQUIRED] list as VAL-01 territory. This is a known open item, not a hidden ambiguity. It is constitutionally preserved as VALIDATION_REQUIRED. Non-blocking.

SUMMARY TABLE

Finding	Status	Blocking	Amendment
F-01A Research/Trade Boundary	REJECTED	NON-BLOCKING	None
F-01B Activities Enumeration	REJECTED	NON-BLOCKING	None
F-02A Halt Term Taxonomy	CONFIRMED	BLOCKING	CREATED
F-02B Governance Halt Exit	CONFIRMED	NON-BLOCKING	None
F-02C Lockout Exit — "Compliance Restored"	CONFIRMED	BLOCKING	CREATED
F-03A Report Cadence	CONFIRMED	BLOCKING	OWNER DECISION REQUIRED
F-03B Report Production Responsibility	REJECTED	NON-BLOCKING	None
F-03C Escalation Report Content	CONFIRMED	NON-BLOCKING	None
F-04A SDM-15 Generic Halt Exit	CONFIRMED	BLOCKING	CREATED
F-04B Hard Deterministic Halt	CONFIRMED	NON-BLOCKING	Subsumed by F-02A
F-04C Open Menu Expiry	CONFIRMED	NON-BLOCKING	None
F-04D Halt Exit Authority Asymmetry	CONFIRMED	NON-BLOCKING	None

Finding	Status	Blocking	Amendment
F-04E Execution Logic Determinism	REJECTED	NON-BLOCKING	None
F-04F Case-by-Case Protocol	CONFIRMED	NON-BLOCKING	None
F-04G Concentration Upper Bound	CONFIRMED	NON-BLOCKING	VAL-REQUIRED

BLOCKING FINDINGS REQUIRING ACTION

ID	Issue	Resolution Path
F-02A	Halt term taxonomy undefined	Constitutional amendment
F-02C	Governance Lockout exit condition undefined	Constitutional amendment
F-03A	Report cadence undefined	Owner decision required before freeze
F-04A	SDM-15 generic halt exit conditions undefined	Constitutional amendment